

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 8.4a: Augmented Generation, RAG vs Fine-Tuning, Applications

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Topic 8.4a Agenda

1. **Augmented Generation**
2. **RAG vs Fine-Tuning**
3. **Economic Applications**
4. **Synthesis: Three Workflow Patterns**

*Arc: T1 Prompting wall → T2 Chunking & embeddings → T3 Vector & graph retrieval → **T4a Augmented generation** → T4b Demo, failures, agentic bridge*

Augmented Generation

From Retrieved Chunks to a Final Answer

- We now have the top- K chunks for the user's query (typically $K = 3$ to 10). What do we do with them?
- **Insert them into a prompt template, send to the LLM.** The model sees:
 1. System instructions (how to behave)
 2. The retrieved chunks (the evidence)
 3. The user question
 4. Output-format instructions (how to cite)
- The LLM now **generates** an answer that is *grounded in the retrieved evidence*.
- This is the only place a generative model is used in the whole pipeline. Everything before this — chunking, embedding, retrieval — exists to put high-quality evidence in front of the model.
- **The single most important design choice** at this stage: the *prompt template*.

Prompt Template — Anatomy

A real RAG prompt has four parts. Lines below are color-coded by role:

SYSTEM:

You are a research assistant for academic economists.
Answer the user's question using ONLY the context below.
If the context does not contain the answer, reply
exactly: "I don't know based on the provided sources."
Cite sources inline using [doc_id:page] notation.

CONTEXT:

[1] (fomc_2024_06:p3)

"Participants observed that inflation remained elevated..."

[2] (fomc_2024_06:p4)

"Several noted that core services prices, especially shelter,
continued to rise..."

[3] (powell_2024_press:p2)

"The Committee remains attentive to the risks that inflation poses

"

Source Attribution Mechanics

- **Why economists need this:** Every claim in an academic paper must be *traceable to a source*. An ungrounded LLM output is not citable. A grounded output is.
- **How to enforce:**
 1. At chunking time, give every chunk a stable doc_id and page (or turn, or section).
 2. At retrieval time, return the metadata along with the chunk text.
 3. In the prompt template, label each chunk [1] (doc_id:page) and instruct the model to cite using that label.
 4. In post-processing, replace the model's inline citations with hyperlinks back to the original document.
- **Before / after example:**
 - *Ungrounded:* "The FOMC was concerned about inflation persistence in 2024."
 - *Grounded:* "The FOMC was concerned about inflation persistence in mid-2024 [fomc_2024_06:p3], particularly in core services prices including shelter [fomc_2024_06:p4]."

Refusal and Faithfulness

- “I don’t know” is a feature, not a bug.
- In an academic context, *fabricating an answer is far worse than refusing to answer*.
- **Faithfulness** (the answer is fully supported by the retrieved context) matters more than *fluency* (the answer reads well). A fluent fabrication is the worst possible outcome for research.
- **How to enforce refusal:**
 - Explicit instruction in the system prompt: “If the context does not contain the answer, reply exactly: I don’t know based on the provided sources.”
 - Set generation temperature low (e.g. 0.0–0.2) to discourage creative inference.
 - In post-processing, run a faithfulness check (T4b) and flag answers whose claims aren’t supported by retrieved chunks.
- **Example refusal (good behavior):**

“I don’t know based on the provided sources. The retrieved minutes from June 2024 do not discuss the impact of fiscal policy.”
- This protects the researcher from quoting fiction.

Worked Example: Full Pipeline Run

Query. *"How did Powell discuss inflation expectations in 2023 press conferences?"*

Pipeline trace:

1. **Embed query** with `bge-large-en-v1.5` → 1024-d vector.
2. **Retrieve top-50** via Chroma + BM25 hybrid. Filter: `doc_type = press_conference` AND `year = 2023`.
3. **Re-rank** the 50 with `bge-reranker-large` → top-5.
4. **Inject top-5 chunks** into the prompt template (previous slide), tagged `[powell_2023_03:p4]`, `[powell_2023_06:p2]`, ...
5. **Send to GPT-4o** at `temperature=0.1`. Receives:
"In 2023 press conferences, Chair Powell repeatedly emphasized that long-run inflation expectations remained well anchored [powell_2023_03:p4], while noting that short-run expectations had risen with realized inflation [powell_2023_06:p2]. By the end of the year, he highlighted that survey- and market-based measures had begun to ease [powell_2023_12:p3]."

RAG vs Fine-Tuning

Decision Matrix: RAG vs. Fine-Tuning

Dimension	RAG	Fine-tuning
Knowledge freshness (update daily)	strong	weak
Source attribution / citations	strong	none
Reduces hallucination	strong	limited
Cost to add new documents	near zero	re-train
Latency at query time	moderate	low
Style adaptation ("write like the Fed")	weak	strong
Output-format adaptation (structured JSON)	limited	strong

Reading the matrix (green = native strength, yellow = workable but limited, red = fundamental weakness): RAG dominates on the four dimensions that matter most for *research* — freshness, attribution, hallucination control, and update cost. Fine-tuning dominates on style and format, which

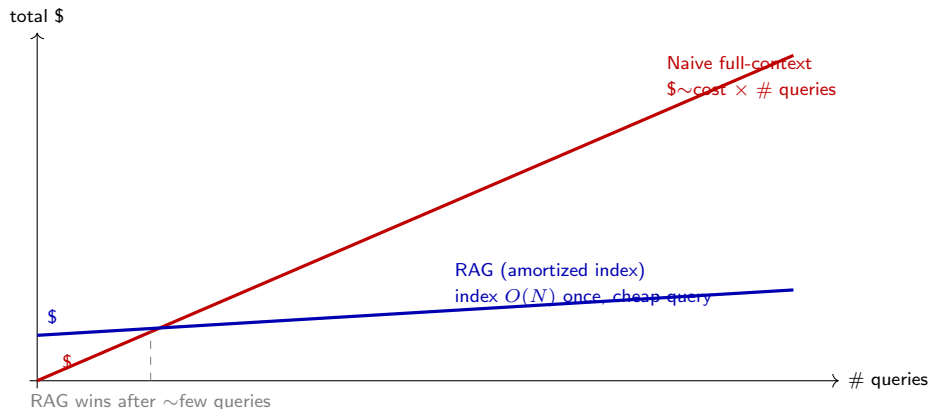
When Fine-Tuning Still Makes Sense

- **Style transfer.** You want the model to *write like a Federal Reserve economist* — the cadence, the hedging, the preferred phrasing. Hand the model a thousand examples of FOMC minutes and fine-tune. RAG can't teach style.
- **Structured-output formatting.** You want the model to always emit a JSON dictionary of DSGE parameters. Fine-tune on ~ 500 input/output pairs. Cheaper than prompt-engineering an enforced schema.
- **Specialized vocabulary.** The model needs to recognize that LSAP = Large-Scale Asset Purchase or that NIPA = National Income and Product Accounts. Some tokens may not even be in the base tokenizer; fine-tuning can teach the model to handle them gracefully.
- **Latency-critical settings.** Fine-tuned models embed knowledge in weights, so they don't need a retrieval round-trip. Useful for real-time applications — but rarely a binding constraint for research.
- **Crucial caveat:** Fine-tuning $\neq$ teaching new *facts*. Models often memorize facts unreliably during fine-tuning and still hallucinate. *If you want the model to know new facts, use RAG.*

The Hybrid Pattern

- In production, the modern pattern is **both, in different roles**:
- **Fine-tune the embedding model** on a domain corpus (say, 100K FOMC sentences with synthetic query-document pairs). This is *cheap* (~\$50 of compute) and improves retrieval quality dramatically.
- **Use a general-purpose LLM as the generator** (Claude, GPT-4o). No fine-tuning of the big model.
- **Use RAG as the glue**. The fine-tuned embedder ensures that the retriever finds the right chunks; the general LLM ensures fluent, faithful generation.
- **Why this works**. You get domain understanding (from fine-tuning) and freshness + attribution (from RAG), without the cost or fragility of fine-tuning a 70B-parameter generator.
- This is the dominant production architecture in 2025 for any LLM application that touches a specialized corpus — legal, medical, financial, academic.

Cost: Index Once, Query Many (Visual)



- **Naive full-context prompting:** cost scales *linearly* with both context size *and* # queries. Every question repays the full corpus token bill.
- **RAG:** pay $O(N)$ embedding calls *once* at index time; each query then ships only the top- K chunks ($\sim 1\text{--}5\%$ of the corpus). Marginal cost per query collapses.

Economic Applications

Use Case 1: FOMC Minutes & Press Conferences

- **Corpus.** The Federal Reserve publishes statements (post-meeting), minutes (3-week lag), press conference transcripts, and (5-year lag) full meeting transcripts. Available 1993–present at `federalreserve.gov`.
- **Index pipeline:**
 1. Scrape PDFs / HTML from the Fed archive.
 2. Parse into speaker turns (minutes) or Q&A pairs (press conferences).
 3. Chunk with rich metadata: `meeting_date`, `doc_type`, `speaker`, `topic_section`.
 4. Embed with `bge-large-en-v1.5`.
 5. Store in Chroma; run BM25 in parallel for hybrid retrieval.
- **Example research queries:**
 - *“How did the Committee's framing of inflation expectations evolve from 2021 to 2024?”*
 - *“Find every appearance of ‘transitory’ in FOMC minutes between 2020 and 2022.”*
 - *“Which participants most often dissented on rate decisions, 2017–2024?”*
- **Reference:** Hansen, McMahon & Prat (2018), *QJE*, “Transparency and Deliberation Within the FOMC: A Computational Linguistics Approach.”

Use Case 2: NBER Working Papers

- **Corpus.** ~32,000 NBER working papers, 1973–present, available at [nber.org](https://www.nber.org). Each has metadata: title, abstract, authors, JEL codes, full PDF.
- **Use case: literature review automation.** Instead of typing keywords into Google Scholar and reading 50 abstracts, ask:
“Find all NBER working papers from 2019 onwards that estimate the slope of the Phillips curve. Return title, authors, year, the central estimate, and the data source.”
- **Pipeline:**
 1. Index titles + abstracts as one collection (cheap, fast, good for keyword screening).
 2. Index full-paper chunks as a second collection (richer evidence for follow-up questions).
 3. Hybrid retrieval over both. Filter on year, JEL_code, author.
 4. Generator outputs a structured table from retrieved papers.
- **Caveat:** Always verify citations against the actual NBER record. Even with RAG, models occasionally garble author lists or year fields.
- **This is one of the highest-value RAG applications for an active researcher.**

Use Case 3: Congressional Record & Legislative Text

- **Corpus.** The Congressional Record contains every speech delivered on the House and Senate floors since 1873. Daily editions are available at `congress.gov`.
- **Use case: tracking how policy debates frame economic issues over time.**
 - “How has the rhetoric around ‘inflation’ shifted in House Banking Committee testimony from 2008 to 2024?”
 - “Which Senators most frequently cite Federal Reserve research in floor speeches?”
- **The structural challenge.** Congressional Record has a deep hierarchy — Congress / session / chamber / date / speaker / bill / section. The T2 chunking gotcha applies in full force. A naive chunker that ignores this hierarchy will produce essentially un-citable retrieval results.
- **Practical fix:** Use a Congress-specific parser (e.g. `congress-api-python`) to extract clean speaker turns with all metadata, then feed those into the chunker.
- **Reference:** Gentzkow, Shapiro & Taddy (2019), *Econometrica*, “Measuring Group Differences in High-Dimensional Choices: Method and Application to Congressional Speech.”

Use Case 4: SEC 10-K Filings (Item 1A Risk Factors)

- **Corpus.** EDGAR hosts ~5,000,000 SEC filings, including the annual 10-K reports of every public US firm. Each 10-K has a standardized “Item 1A: Risk Factors” section disclosing material risks.
- **Use case: cross-firm comparison of disclosed risks.**
 - *“Compare how the four largest US banks discussed interest-rate risk in their 2022 vs. 2023 10-K filings.”*
 - *“Find all 10-K filings from energy firms in 2024 that mention transition risk from climate policy.”*
- **Pipeline:**
 1. Use `sec-edgar-downloader` to pull 10-Ks.
 2. Run a section-aware parser to extract Item 1A only.
 3. Chunk *within* Item 1A, preserving `cik`, `ticker`, `filing_date`, `fiscal_year`.
 4. Store in Chroma. Hybrid retrieval with metadata filter on `ticker` for cross-firm queries.
- **References:** Loughran & McDonald (2011), *JF*, “When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks.” Lopez-Lira (2023), *working paper*, “Risk Factors That Matter.”

Use Case 5: Cross-Country Central Bank Speeches

- **Corpus.** The Bank for International Settlements maintains a unified archive of central bank speeches from ~60 institutions, 1997–present, at bis.org/cbspeeches. Other primary sources: ECB, BoE, BoJ, RBA, BoC, RBI, PBoC.
- **Use case: comparative monetary policy analysis.**
 - *“How did major central banks frame the post-COVID inflation surge in 2022 speeches? Compare the Fed, ECB, BoE, and BoJ.”*
 - *“Which central banks most often cite financial-stability concerns alongside inflation, 2010–2024?”*
- **Pipeline notes:**
 - Add country, institution, speaker_role, language as metadata.
 - For non-English speeches, either use a multilingual embedder (multilingual-e5-large) or translate first.
 - Hybrid retrieval is critical — many speeches use country-specific institutional jargon a generic embedder won't recognize.
- **This use case is essentially impossible without RAG — no human can read ~10,000 speeches across ~60 institutions per year.**

Use Case 6: Combining Structured + Unstructured

- RAG isn't only for unstructured text. The most powerful pattern combines a **structured database** (e.g. FRED time series, BEA NIPA tables) with an **unstructured text store** (Fed speeches, FOMC minutes).
- **Example query:**
"In months when the unemployment rate fell below 4%, what did the Fed publicly say about labor markets?"
- **Pipeline:**
 1. *Structured side:* SQL on FRED data. `SELECT date FROM unrate WHERE value < 4.0;`
 2. *Unstructured side:* retrieve speeches and FOMC minutes whose date falls within those months.
 3. Pass the joined evidence to the LLM with a prompt that says "summarize how Fed officials characterized the labor market in these specific months."
- **This pattern** — structured SQL filter → semantic retrieval → generation — is uniquely useful for empirical economists. It bridges what economists already know how to do (data) with what they couldn't do before (text at scale).

Synthesis

Three-Roles Taxonomy: What Is the RAG Output For? (Callback to Lec. 7 T4)

Use case	Signal	Outcome	Instrument
FOMC minutes	FOMC stance over time	asset prices, expectations	communication shocks
NBER papers	methodological frontier	citation counts	field-maturity proxy
Congressional Record	policy rhetoric	bill passage, vote share	constituent preferences
SEC 10-K	firm-disclosed risk	stock returns, default	risk-disclosure rule shock
Central-bank speeches	cross-country stance	exchange rates	comparative monetary shock

Reading the table. RAG returns *evidence*; the econometric role you assign that evidence determines *how you use it*. The same five corpora support **signal** extraction (text-as-variable), **outcome** measurement (text-as-target), or **instrument** construction (text-as-IV). Lec. 7 T4 introduced the taxonomy on the LLM side; the same three roles transfer one-to-one to RAG outputs.

Synthesis: Three Workflow Patterns

1. **Literature review.** Index a paper repository (NBER, SSRN, RePEc). Query: “find work on X, summarize, return citations.” Saves days per project.
2. **Policy-text monitoring over time.** Index an institutional corpus (FOMC, Congress, central banks). Query: “how did the framing of Y evolve from year A to year B?” Turns institutional text into a longitudinal research instrument.
3. **Cross-sectional comparative analysis.** Index a corpus where the unit of analysis is a firm or country (10-Ks, central bank speeches). Query: “compare how entity-1, entity-2, . . . discussed Z in year Y.” Enables cross-firm and cross-country comparisons that were previously infeasible at scale.

Methodological punchline. RAG turns institutional text from *background reading* into a *queryable research instrument*. That is the new capability. Everything else — chunking, embeddings, vector stores — is plumbing.

Topic 8.4a Wrap-Up

- **What changed** once we added augmented generation: retrieved chunks become *evidence*, and the generator is forced to cite them. The output is now *auditable*.
- **When to reach for fine-tuning instead**: style, format, jargon — not facts. For facts, RAG always wins.
- **Five corpora, six pipelines**: FOMC, NBER, Congress, SEC 10-K, BIS speeches, structured + unstructured. Same plumbing; different research design.
- **Methodological move**: institutional text becomes a queryable research instrument. The Section IV synthesis is the take-home.
- **Up next (Topic 8.4b)**: a working OLG demo you can run today, the six common RAG failure modes, evaluation with RAGAS + Cohen's κ , and the explicit bridge to agentic RAG in Lecture 9.